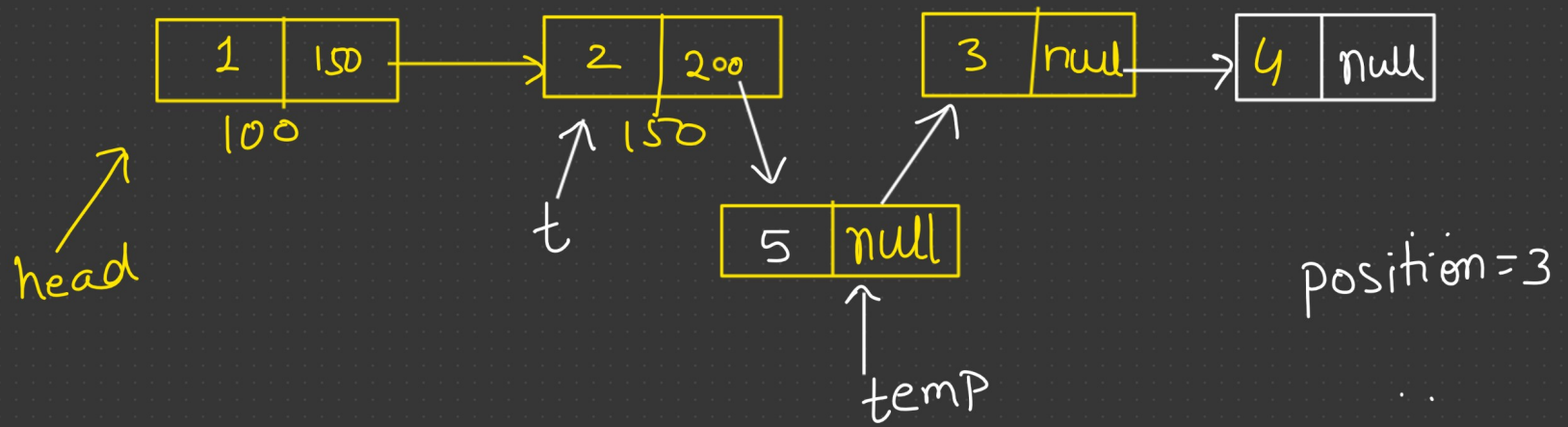


insert\_at\_end(int value)

```
{  
    node* temp = new Node();  
    node* t = head;  
    while (t → next != NULL)  
    {  
        t = t → next;  
    }  
    t → next = temp;  
    temp → data = value;  
}
```

```
if (head == null)  
{  
    head = temp;  
}
```



```
node *temp = new node();
temp->data = value;
```

```
node *t = head;
```

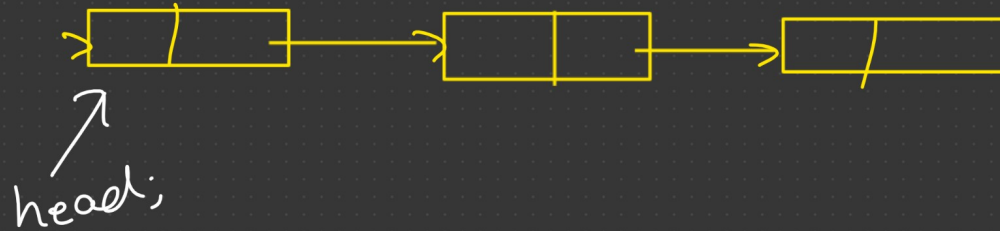
```
while ((position-2) != 0)
{
    t = t->next;
    position--;
}
```

```
temp->next = t->next;
t->next = temp;
```

```
if (position == 1)
{
    temp->next = head;
    head = temp;
}
```

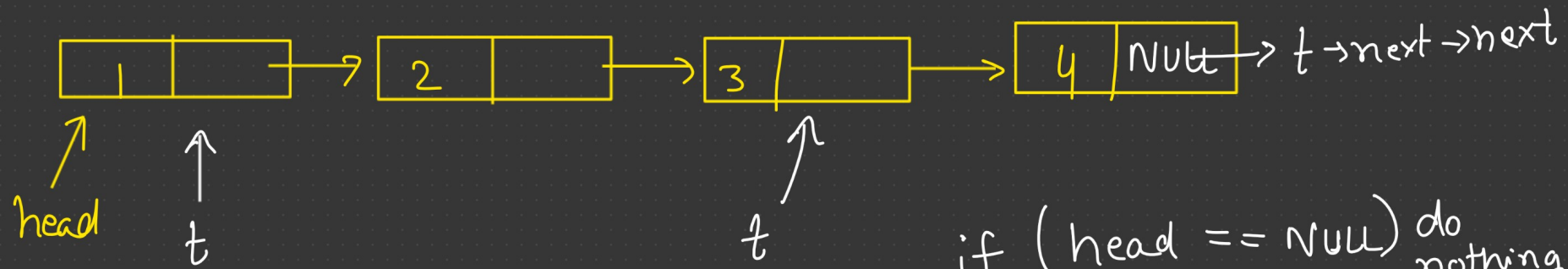
## Deletion

temp  
↓



```
if (head == null)
    cout << "Not possible";
```

```
temp = head;
head = head->next;
delete (temp);
```



```
while (t->next->next != null)
```

```
    t = t->next;
```

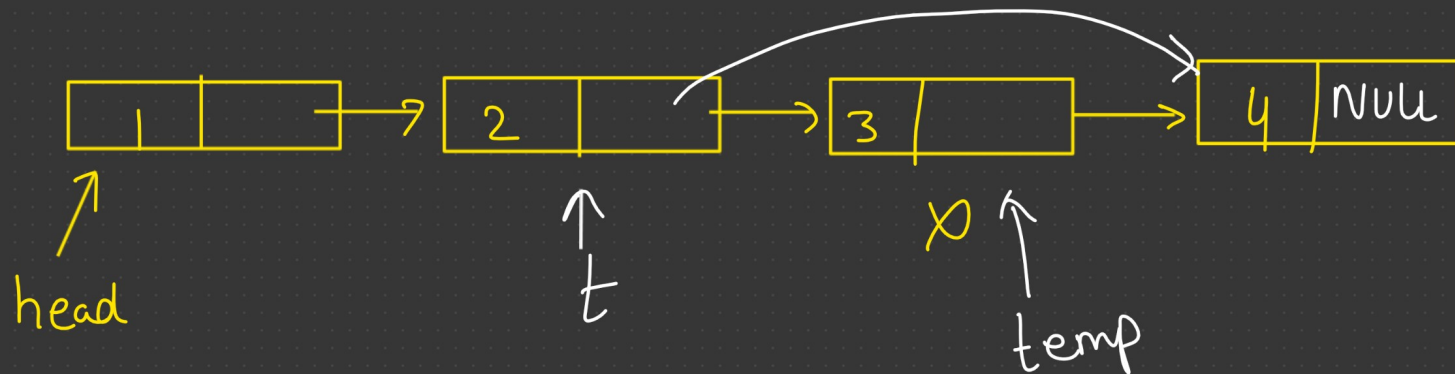
```
    delete(t->next);
```

```
    t->next = null;
```

```
if (head == NULL) do nothing
```

```
if (head->next == NULL)
```

```
    head = NULL;
```



```
while ((position - 2) != 0)
```

```
    t = t->next;
```

```
    node* temp = t->next;
```

```
    t->next = temp->next;
```

```
    delete(temp);
```

```
if (head == NULL)
```

```
    not possible
```

```
if (position == 1)
```

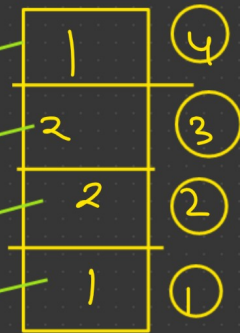
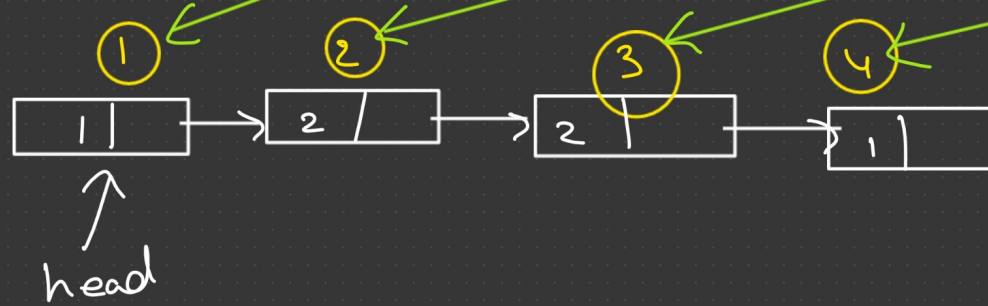
```
    t = head;
```

```
    head = head->next;
```

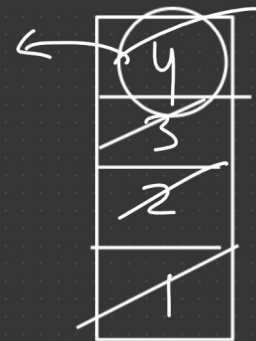
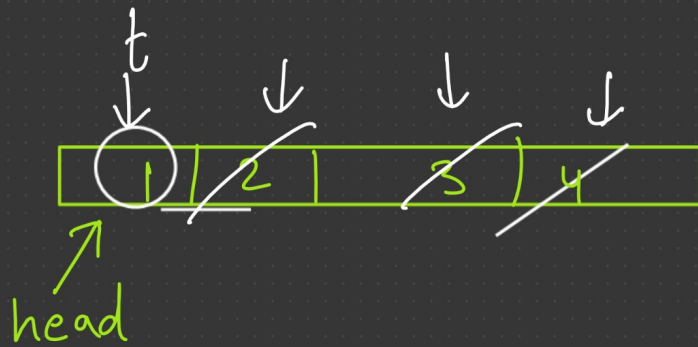
```
    delete(t);
```

# Palindrome:-

Stack <int> s;



LIFO





```
bool isPalindrome( *head )
```

```
{
```

```
    stack<int> S;
```

```
    ListNode *temp = head;
```

```
    while (temp != NULL)
```

```
    {
```

```
        S.push( temp->val );
```

```
        temp = temp->next;
```

```
    }
```

```
    temp = head;
```

```
    while( temp != NULL)
```

```
    {
```

```
        if ( temp->val != S.top() )
```

```
            return false;
```

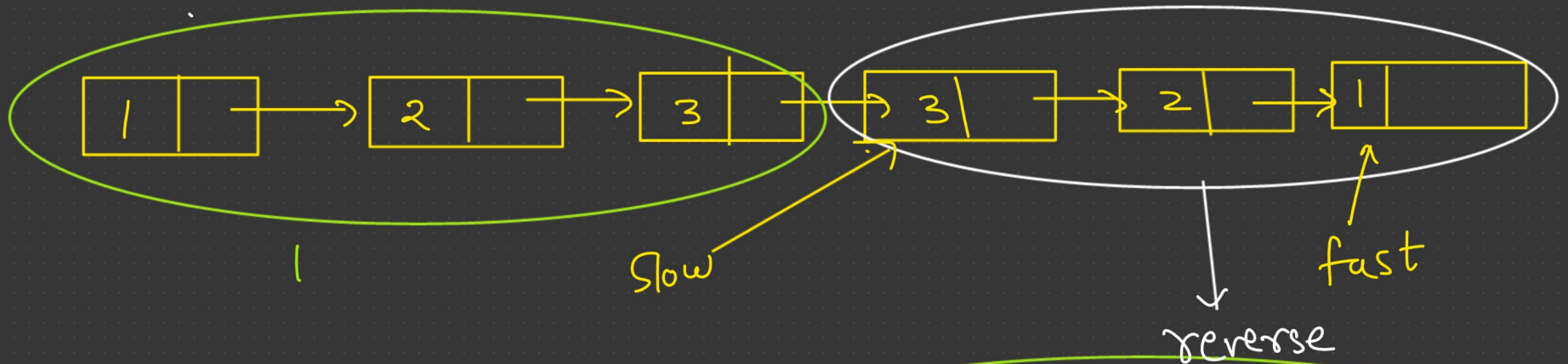
```
        temp = temp->next;
```

```
        S.pop();
```

```
    }
```

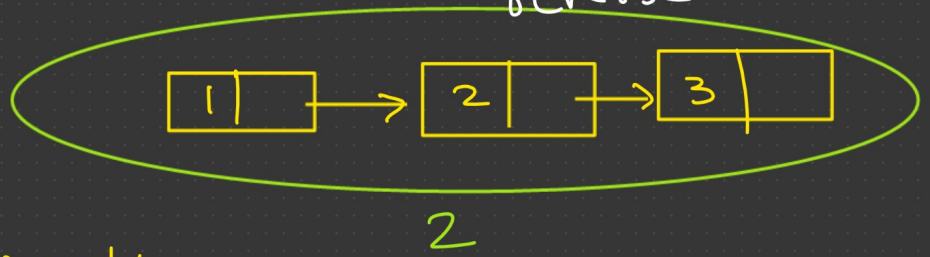
```
    return true;
```

```
}
```



slow = slow → next;

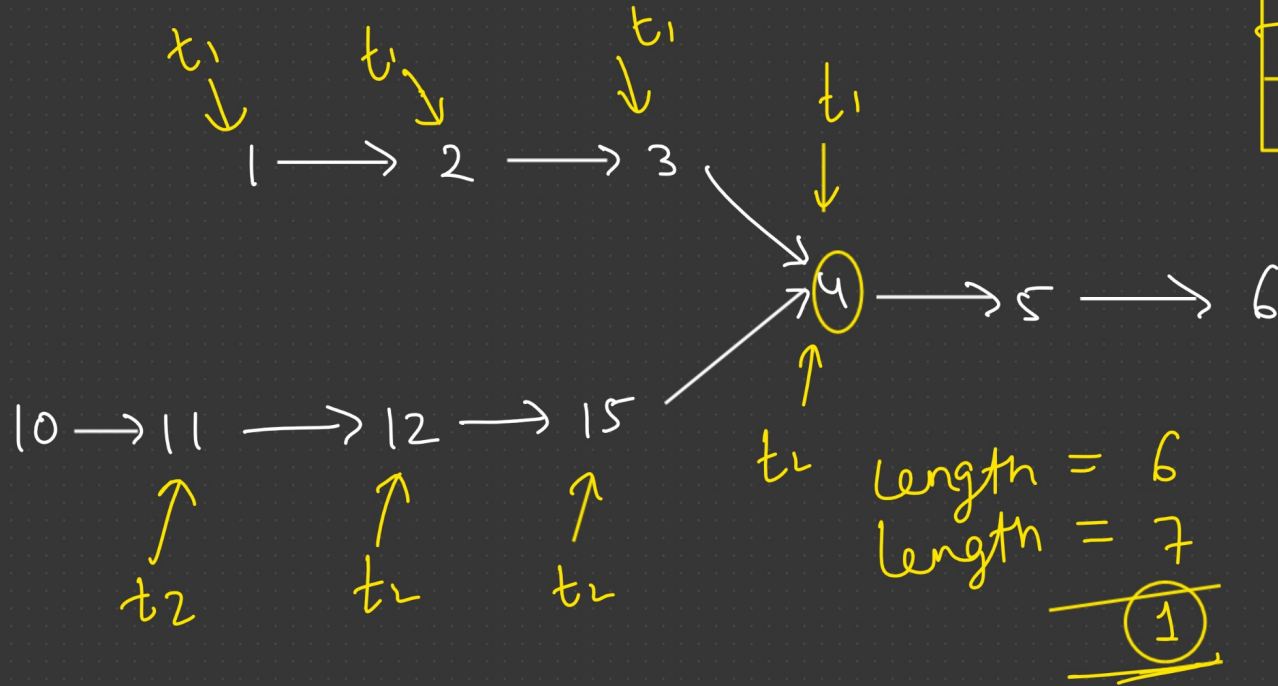
fast = fast → next → next;





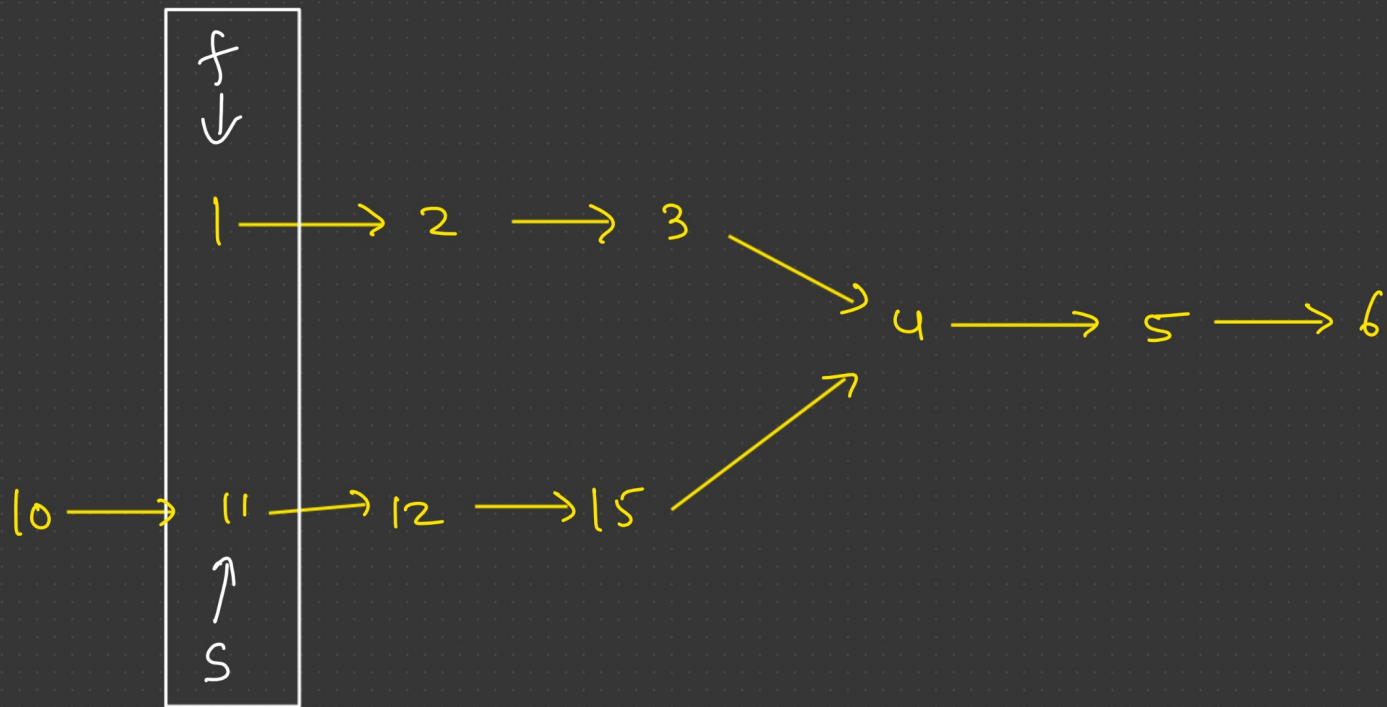
Ans = ~~6~~  
~~5~~ (4) ✓

### Intersection:-



|              |
|--------------|
| <del>6</del> |
| <del>5</del> |
| <del>4</del> |
| 3            |
| 2            |
| 1            |
| $S_1$        |

|              |
|--------------|
| <del>6</del> |
| <del>5</del> |
| <del>4</del> |
| 15           |
| 12           |
| 11           |
| 10           |
| $S_2$        |



① Singly list

② Doubly ll

③ Circular

④ LeetCode -

H.W

Hashnode

Article